

Solution Architecture

Date: 29 September 2026

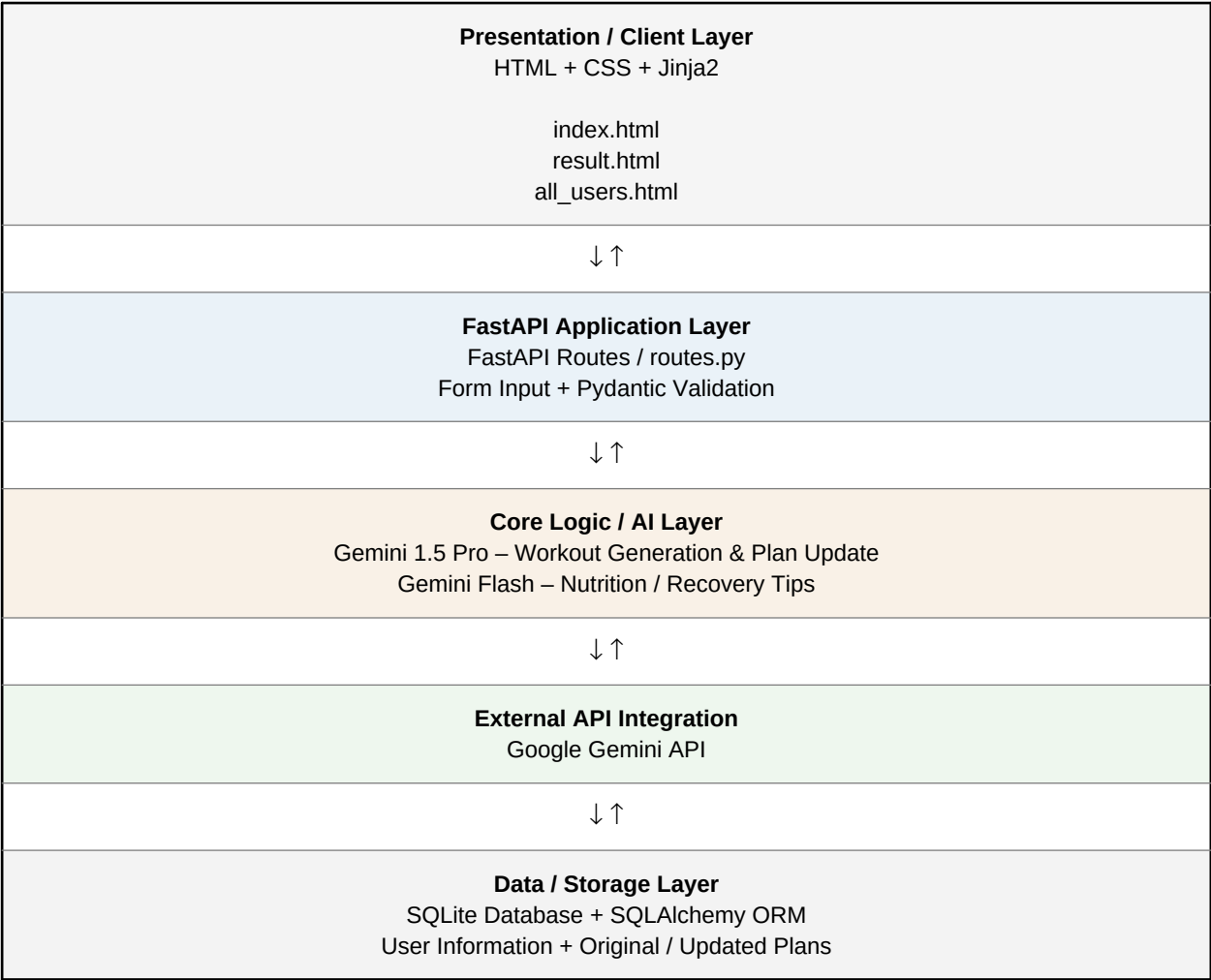
Team ID: _____

Project Name: FitBuddy – AI Fitness Plan Generator using Gemini Models

Maximum Marks: 5 Marks

Solution Architecture Diagram:

A high-level architecture of FitBuddy, showing the presentation, application/AI, external API, and data/storage components.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Collects user details, displays the personalized 7-day workout plan, nutrition/recovery tip, feedback form, and admin user view.	HTML, CSS, Jinja2
API / Application Layer	Receives and validates user input, manages routes, connects frontend with AI functions, and returns generated results.	FastAPI, Python, Pydantic
Core Logic / AI Layer	Generates structured workout plans, updates plans using feedback, and provides quick nutrition or recovery tips.	Gemini 1.5 Pro, Gemini Flash
External API Integration	Connects the application to Google Gemini models for AI-powered workout generation, plan updating, and nutrition tips.	Google Gemini API, google-generativeai
Database	Stores user information and AI-generated plans. Keeps original and feedback-updated plans for later viewing.	SQLite, SQLAlchemy ORM
Server / Deployment	Runs the FitBuddy application locally and serves the web interface and API endpoints.	Uvicorn, Python virtual environment

Architecture Flow: User → HTML/Jinja2 Interface → FastAPI → Gemini AI → Generated Workout/Nutrition Result → SQLite Storage. Feedback is sent through FastAPI to Gemini 1.5 Pro to update the existing plan.

The architecture is based on the FitBuddy project description and implementation details, including FastAPI, Gemini models, Jinja2, and SQLite/SQLAlchemy. [filecite turn2file0 L98-L114](#)